

HAT-P-19b

R-1053

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_241203 · created 2026-07-30T10:18:19+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460647.7108 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.025
Transit depth (Rp/Rs)^2	2.69 +/- 0.83 [%]
Orbital Inclination (inc)	88.48 +/- 0.79
Airmass coefficient 1 (a1)	0.99 +/- 0.015
Airmass coefficient 2 (a2)	0.006 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.49 %
Best Comparison Star	#3 - [237, 426]
Optimal Method	PSF photometry
Transit Duration (day)	0.0953 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.164 ± 0.025 vs 0.1418 (deviation 0.8σ)
Duration fit vs published	0.0953 d vs 0.1196 d (deviation 3.96σ)
Mid-time O-C	3.0 min
Depth SNR	5.9
Residual scatter	1.49 %

Light curve

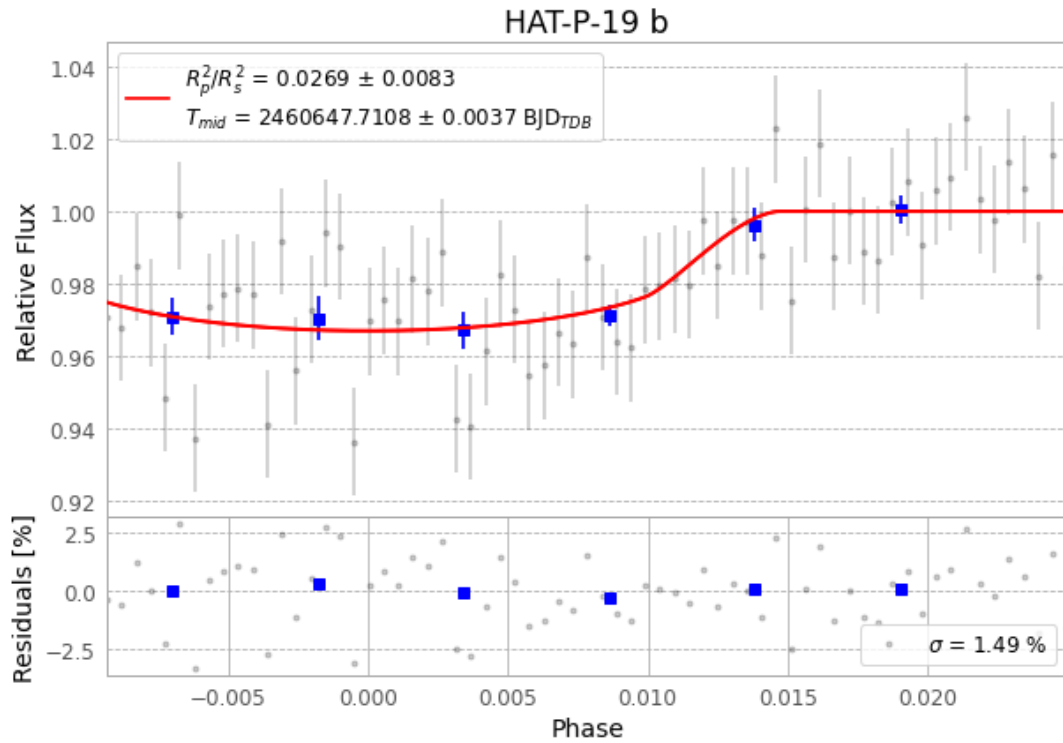


Figure 1 — FinalLightCurve_HAT-P-19 b_2024-12-02.png (SHA-256 6246f298a0345e71...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 285], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 02461a13f0472aeb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), HATP-19241203040315.FITS → HATP-19241203072115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-241203.log (bdfeca7bdc97...), FinalLightCurve_HAT-P-19 b_2024-12-02.png (6246f298a034...), FinalLightCurve_HAT-P-19 b_2024-12-02.csv (37d177d5c51c...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 10:18Z.